



FIRST SEMESTER 2026-2027
Course Handout (Part II)

Date: 11/08/2026

In addition to part I (General Handout for all courses appended to the timetable), this portion gives further specific details regarding the course.

Course No. : MATH U101

Course Title : Multivariable Calculus

Instructor-in-charge: PRAMOD EYYUNNI (6070-G, pramod.eyyunni@pilani.bits-pilani.ac.in)

Team of Instructors : Balram Dubey, Devendra Kumar, Sangita Yadav, Gaurav Dwivedi, Trilok Mathur, Shailesh Trivedi, Jitender Kumar, Divyum Sharma, Rahul Kumar, Dhiraj Kumar Das, Shuchita Goyal, Sudarshan Santra, Deepak Bhorliya

1. Course Description:

This course provides a foundational introduction to multivariable calculus and vector analysis, which are essential tools in engineering and the physical sciences. Students learn to analyze functions and geometric objects in two and three dimensional spaces, studying local behavior through differentiation and global properties through integration. Compared to single variable calculus, multivariable calculus introduces new ideas and describes behaviour in multiple directions and dimensions. These concepts play a central role in areas such as probability and statistics, fluid mechanics, electrical sciences and numerous other applications.

The course develops mathematical techniques required to model real-world phenomena such as motion, flow, and fields. Key topics include polar coordinates, convergence of sequences and series, power series, Taylor and Maclaurin series, partial derivatives, and vector calculus, leading to fundamental integral theorems such as those of Green, Stokes, and Gauss.

2. Scope and objectives of the course:

- To develop a solid conceptual understanding of functions of several variables.
- Introduce partial derivatives and their applications in modelling and optimization.
- To study multiple integrals and their use in computing areas, volume and mass.
- Build geometric intuition via visualization of surfaces and vector fields.
- To strengthen problem solving skills required for advanced engineering courses.

3. Course learning outcomes:

Upon successful completion of this course, students should be able to:

- **CLO1.** Determine if an infinite series is convergent or divergent by selecting the appropriate tests.
- **CLO2.** Apply techniques for plotting and analyzing curves in polar coordinates, including converting between polar and Cartesian forms, sketching polar graphs.





- **CLO3.** Interpret parametric curves as trajectories described by position vectors, and compute associated velocity and acceleration vectors to analyze motion.
- **CLO4.** Develop a deep understanding of the gradient vector, including its geometric meaning, its relationship to level curves and level surfaces, its role in directional derivatives, and its use in linear approximation and sensitivity analysis.
- **CLO5.** Compute partial derivatives using the chain rule and total differentials, and apply these tools to model changes in multivariable systems.
- **CLO6.** Formulate and solve optimization problems involving multiple variables, both unconstrained and constrained, using techniques such as Lagrange multipliers.
- **CLO7.** Compute multiple integrals using appropriate coordinate systems, including rectangular, polar, cylindrical, and spherical coordinates.
- **CLO8.** Perform change of variables in multiple integrals and understand the role of the Jacobian determinant.
- **CLO9.** Set up and evaluate line integrals (for work and circulation), surface integrals (for flux), and volume integrals, and interpret their physical significance in applied problems.

4. Method of conducting the course:

The instructional approach for this course integrates theory, practice, and computational exposure. Core concepts and mathematical derivations will be introduced through classroom lectures, while tutorial sessions will be devoted to reinforcing these ideas through guided problem-solving and application-based exercises. To supplement in-class learning, three video lectures will be made available, showcasing the use of computational software (MATLAB) for visualization and problem-solving. Student learning will be evaluated on a continuous basis through quizzes, assignments, and examinations, ensuring both conceptual understanding and practical competence.

5. Textbook:

TB. J. Hass, C. Heil, P. Bogacki and M. D. Weir: *Thomas' Calculus*, 15th Edition, Pearson Education, 2024.

6. Reference Books:

- RB1.** E. Kreyszig : *Advanced Engineering Mathematics*, 10th Edition John Wiley and Sons, 2011.
- RB2.** T. M. Apostol : *Calculus Vols I and II*, 2nd Edition, John Wiley and Sons, 1967 and 1969.
- RB3.** J. Stewart: *Essential Calculus Early Transcendentals*, Thomson Learning, 2014.
- RB4.** S. Lang: *A First Course in Calculus*, Springer-Verlag 5th edition, 2009.
- RB5.** R. L. Lipsman, J. M. Rosenberg: *Multivariable Calculus with MATLAB® With Applications to Geometry and Physics*, Springer-Verlag, 1st edition, 2017.





7. Course Plan: Section numbers below are from the textbook TB, unless otherwise specified.

Module	Topics	Section No.	Learning Outcome
1.	Limits and continuity of a real-valued function of one real variable L1: Convergence of sequences and series of real numbers L2-L4: Different tests of convergence for the series of non-negative terms, absolute and conditional convergence, alternating series L5: Power series, radius of convergence L6: Maclaurin and Taylor series of functions	2.3-2.6** (Self-study) 10.1-10.8	Revision of Class 12 topics essential for this course Differentiate clearly between the three types of series convergence with examples and counterexamples, approximating functions with polynomials
2.	L7-L8: Polar coordinates, graphing polar functions of the form $a \pm b \sin \theta$ or $a \pm b \cos \theta$	11.3 -11.4*	Understand the curvilinear coordinate systems, which can be more natural than Cartesian coordinates sometimes
3.	L9: Limit, continuity, and differentiability of vector functions L10: Arc length, velocity, unit tangent vector L11: Curvature, normal vector	13.1, 13.2*, 13.3, 13.4	Understanding of vector-valued functions, motion, and spatial paths, relation between the dynamics and geometry of motion
4.	L12: Functions of several variables, level curves L13: Limits, continuity of two/three variable functions	14.1, 14.2	Understanding of multivariable limits and continuity, which are more intricate than the single-variable case
5.	L14: Partial derivatives L15: Differentiability, chain rule	14.3, 14.4	Difference between ordinary and partial derivatives
6.	L16: Directional derivatives, gradient vectors L17: Tangent planes, normal lines, linearization, total derivative	14.5, 14.6*	Generalizations of partial derivatives and their applications





7.	L18-L20: Extrema (Maximum, minimum, and saddle points) L21: Lagrange multipliers	14.7, 14.8	Optimization (maximize or minimize) functions of several variables locally as well as globally
8.	L22-L23: Double integrals, areas L24-L25: Change of integrals from rectangular to polar coordinates	15.1-15.4	Evaluation of area and volume using iterated integrals
9.	L26-L28: Triple integrals in rectangular, cylindrical, and spherical coordinates L29-L30: Substitution in integrals	15.5, 15.7, 15.8	Volumes of solids in space using a suitable curvilinear coordinate system
10.	L31: Line integrals L32-L33: Work, circulation, flux, path independence, potential functions, conservative fields L34-L35: Green's theorem in the plane	16.1-16.4	Different integrals of vector fields on objects in space; applications to flow, flux, work, etc.; their mutual relationship via Green's theorem generalizing the fundamental theorem of integral calculus
11.	L36-L37: Surface area and surface integrals L38-L40: Gauss' divergence theorem, Stokes' Theorem	16.5-16.8*	Divergence and Stokes' theorems as further generalizations of Green's theorem
12.	Video lectures to be uploaded on Nalanda for demonstration of MATLAB	Self-study** (Reference Book – RB5)	Introduction of MATLAB for basic arithmetic operations and computations, generate and visualize 2D and 3D plots, manipulate 3D plots, including rotating and viewing surfaces from different perspectives, interpret changes in different directions using graphical tools, symbolic computations using the MATLAB Symbolic Toolbox.

Notes:

- * - Exclude lengths of polar curves and graphing other polar curves (Sec 11.4), projectile motion (Sec. 13.2), error estimation (Sec. 14.6) and laws of electromagnetic theory and hydrodynamics (Sec. 16.8).
- ** - Self-study sections: 2.3-2.6, Video lectures





8. Evaluation Scheme:

S. No.	Evaluation components	Weightage (Marks)	Duration (Minutes)	Date & Time	Nature of component (Closed Book/ Open Book)
1	Mid-semester exam	30% (90)	90	As announced by AUGSD	Closed book
2	Quizzes* (4 Announced quizzes, to be held in the respective tutorial sections)	20% (60)	20 minutes each	To be announced by the instructors of the respective tutorial sections	Closed book
3	Comprehensive exam	45% (135)	180	As announced by AUGSD	Closed book + Open book
4	Assignment (based on video lectures)	5% (15)	---	To be announced by IC	Take-home

***There will be four announced quizzes of 20 marks each, out of which best three will be considered.**

9. Make-up Policy: Make-up for mid-semester and comprehensive exams will be given only for very genuine cases, and prior permission must be obtained from IC. **No makeup will be given for quizzes under any circumstances whatsoever.**

10. Attendance Policy: Each student is expected to attend all classes and appear in all evaluative components.

11. Cut off for NC grade: Any student scoring less than 10% of the total marks (i.e., less than 30 marks out of 300) will be given NC grade. Further, if a student is absent in mid-semester or comprehensive exam or in all four quizzes and assignment submission, then he/she will be given NC grade.

12. Chamber consultation hour: To be announced by the respective instructor. The chamber consultation hours of all the instructors will be uploaded on the Nalanda website.

13. Notices: The notices concerning this course will be displayed on the Nalanda website.

**Instructor-in-charge
MATH U101**

